

ShopSmart	
Project Plan	Date: 2024/03/15

ShopSmart Project Plan

1 Introduction

In this document, we will outline our project plan for ShopSmart. ShopSmart is an easy to use, review platform that connects customer and sellers where the sellers can publish their products and customers give ratings and reviews and have a discussion over a public forum. This document will consist of our planned project organization, development practices, planned milestones and estimated delivery dates, deployment practices. At the end of this document, we will share the insights we have gained during the preparation of this document.

2 Project organization

Our project contains the following members:

- **Furkan Şimşekli**, our **project manager** who will:
 - Determine our development process model, which will be included in a later section in this document
 - Split our goals into atomic parts, who will then assign those parts to other project members
 - Track the progress of the parts and re-assign the tasks if necessary to other members to meet the deadlines
 - Take part in coding the backend and frontend during project
- **İlker Avcı**, our **software architect** who will:
 - Decide which software stack the project group will use
 - Design the outline of the software architecture
 - Determine the software requirements and constraints before and throughout the development
 - Involve in the hands-on elements of software delivery, and provide guidance to other developers
 - Take part in coding the backend during project
- **Mete Uysal**, our **software tester** who will:
 - Prepare unit and integration tests against the codebase
 - Present the test results to development team at our recurring meetings
 - Improve the end product by applying QA tests throughout the project life cycle
 - Take part in coding the backend and frontend during project
- **Eyüp Menevşe**, our **software analyst** who will:
 - Ensure the quality and efficiency of the code written by the development team
 - Determine the potential risks which can be encountered
 - Make sure the team adheres to the requirements and desire of stakeholders
 - Take part in coding the backend and frontend during project, design the UI/UX
- **Buğra Kağan Acar**, our **software configuration manager** who will:
 - Manage the development environment and tooling
 - Determine the procedures which will be followed during the usage of version control systems

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- Define versioning schema, baselining, build and release procedures
- Handle the dependency management and supply chain issues
- Take part in coding the backend during project

Team Member	Software Developer	Software Architect	Software Project Manager	Software Tester	Software Analyst	Software Configuration Manager
Buğra Kağan Acar	X					X
Eyüp Menevşe	X				X	
Furkan Şimşekli	X		X			
İlker Avcı	X	X				
Mete Uysal	X			X		

3 Development process and measurements

We will be using an iterative development process, specifically a variant of Agile methodology. This allows for continuous feedback and adaptation throughout the development lifecycle. The project will be broken down into a series of time-boxed iterations, usually 1 or 2 weeks length sprints. Each iteration will focus on delivering a specific set of features and functionalities outlined in a product backlog.

Planning: The team will meet at the beginning of each iteration to prioritize tasks from the product backlog, estimate effort, and define exact sprint goals by looking at the initial project planning. Tasks will be shared in a way, so that dependency between the developers will be minimum since the team members mostly have different schedules.

Development: Developers will work on assigned tasks, implementing features and functionalities according to the sprint plan. Developers will work on their respective branches based on the feature or the fix they are working on. Linters and formatters from GitHub workflow will run after committing a change into codebase in order to keep the codebase clean and understandable for other members of the team.

Testing: Testers will continuously test throughout the iteration to identify and fix bugs. Unit tests, integration tests will be conducted. If needed, user acceptance tests can be also conducted by releasing beta versions at the later stages of the project.

Stand-up Meetings: Short meetings will be held to keep everyone informed and updated about progress, identify roadblocks, and adjust plans as needed. The frequency of these meetings will depend on the workload of the people who are committing into this project, preferably 2 or 3 times a week.

Code Review: Every piece of code written by developers will be reviewed by peers to ensure quality and adherence to coding standards.

Deployment: At the end of each iteration, a potentially shippable product increment will be deployed to a staging environment for internal testing and review. In case of immediate need, hotfixes can be done at the production to see the results quickly.

We will utilize a CI/CD pipeline to automate the build, test, and deployment process. This will allow for frequent integration of code changes and faster deployments with minimal risk.

The team will use Git to track code changes, maintain different versions of the software, and enable collaboration by utilizing branches, issue trackers on GitHub.

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4 Project milestones and objectives

Project milestones have been given on the table below. Although the whole picture isn't crystal clear yet, we can assume the schedule will be like the given below for now. On our way until the end of deadline, it's possible to make adjustments on the objectives, and the target velocity.

Please note that full list of use cases haven't been defined at this stage of the project, therefore they are not included in objectives. On our way, they will be added under the module implementation objectives. Also, it's important to notice that only start dates have been given at the table. Our working process heavily relies on the feedback that we get from our own work, and estimation of the due dates is unclear at this stage. However, in our development environment, each task we are going to implement will be bounded to releases, and releases will have an estimated date to publish.

DTD stands for "Develop, Test, Deploy".

Phase	Iteration	Primary objectives	Scheduled start	Target velocity
Project Init	I0	1. Identify project constraints 2. Defining system functionalities 3. Set roles within the team 4. Set recurring communication channels and meetings within the project team	2024.03.01	12
Project Init	I1	1. Outline the module relations and high level software design 2. Develop Project Plan 3. Develop Project Vision	2024.03.08	21
Project Init	I3	1. Identify and outline project requirements 2. Define use case scenarios 3. Define system-wide requirements 4. Define test cases	2024.03.15	24
DTD	I0	1. Design database models and relations 2. Develop Risks Management 3. Build the production environment 4. Develop Configuration/Change Management	2024.03.22	36
DTD	I1	1. Implement the authentication system 2. Implement the Administrator Module 3. Run unit tests for Administrator	2024.03.29	42

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		Module		
DTD	I2	1. Implement the User Module 2. Run unit tests for User Module 3. Integrate Administrator Module and User Module	2024.04.05	42
DTD	I3	1. Present demo, and evaluate the given feedbacks 2. Implement the Merchant Module 3. Run unit tests for Merchant Module	2024.04.19	38
DTD	I4	1. Integrate Merchant Module to rest of the modules 2. Implement Community Management Module	2024.04.26	27
DTD	I5	1. Integrate Community Management Module to rest platform	2024.05.10	21
Final	I1	1. Simulate edge-case use and security scenarios 2. Do a checklist of all use cases 3. Update Risks Management 4. Update Configuration/Change Management	2024.05.17	33
Final	I2	1. Present the project	2024.05.24	15

5 Deployment

We will employ a continuous integration system, that checks the current revision against a set of tests and deploy the system to a server if the tests pass using a continuous deployment system. The aforementioned server will be a rented Linux server that'll be automatically scanned for issues.

6 Lessons learned

During the project planning, our team recognized the importance of fostering a conducive working environment that encouraged open dialogue and collaboration among team members. Acknowledging that each team member brings unique skills and perspectives to the table, we realized the value of leveraging the diversity.

Consequently, we maintained flexibility in our approach. As we progressed, we observed how decisions made during our initial meetings could evolve significantly during subsequent discussions. This experience underscored the potential pitfalls of treating initial assumptions as immutable truths. We learned the significance of allowing room for error and course correction as the project unfolded.